

Muhammad Sajid

Full-Stack Web Developer

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PROFESSIONAL SUMMARY

Full-Stack Web Developer and Software Engineering student at MUET with hands-on experience architecting and shipping scalable MERN-stack applications. Delivered real-world projects spanning AI-powered platforms, real-time systems, and geospatial data intelligence each featuring secure JWT authentication, RESTful APIs, RAG-based AI pipelines, and polished responsive UIs. Passionate about clean architecture, performance optimization, and user-centered design.

TECHNICAL SKILLS

Frontend: React.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, EJS, Leaflet, Recharts

Backend: Node.js, Express.js, RESTful API Design, Socket.IO, JWT Authentication, Redis (Upstash), BullMQ, Server-Sent Events (SSE)

Databases: MongoDB, MySQL, Mongoose, Pinecone (Vector DB)

AI & Data: Google Gemini API/SDK, Groq SDK, Retrieval-Augmented Generation (RAG), DBSCAN Clustering

Tools & DevOps: Git, GitHub, Vite, VS Code, Docker, Prometheus, Pino, Cloudinary, Google Earth Engine, Resend

Concepts: MVC Architecture, Role-Based Access Control, Responsive Design, OOP

PROJECTS

NovaMind AI — Enterprise-Grade Full-Stack AI Chatbot Platform [\[Live\]](#)

June 2026

React (Vite) · Tailwind CSS · Node.js · Express.js · MongoDB · Redis (Upstash) · Google Gemini API · Pinecone · BullMQ · Cloudinary · Docker · Prometheus · JWT

- Architected a production-hardened AI chatbot platform with multi-model Gemini key rotation, automatic fallback chains, and real-time token streaming via Server-Sent Events supporting stop-generation and message versioning.
- Engineered an asynchronous RAG ingestion pipeline using BullMQ and Upstash Redis to parse, chunk, embed, and upsert PDF/DOCX/XLSX/PPTX files into Pinecone, powering context-aware responses over user-uploaded documents.
- Built a JWT dual-token authentication system with httpOnly cookies and OTP email verification via Resend, alongside Prometheus telemetry and Pino structured logging, deployed via Docker on Render and Vercel.

ThermaX — Hybrid Crowdsourced Urban Heat Intelligence System (Final Year Project) [\[GitHub\]](#)

Jan 2026 – Present

React (Vite) · Tailwind CSS · Node.js · Express.js · MongoDB · DBSCAN · Leaflet · Google Earth Engine

- Engineered a full-stack civic intelligence platform fusing GPS-tagged citizen heat reports with NASA/MODIS satellite imagery via Google Earth Engine to detect Urban Heat Island hotspots across Pakistani cities.
- Implemented a DBSCAN-powered spatial clustering pipeline with a custom heat score formula and StandardScaler preprocessing to identify high-risk zones without requiring a predefined cluster count.
- Built an interactive GeoPulse dashboard with a live Leaflet heatmap, hotspot polygons, clustered markers, and Recharts analytics, with role-based access for reporters, analysts, and admins.

Preparation Mate — AI-Powered Study & Career Preparation Platform [\[GitHub\]](#)

July 2026 – Present

React (Vite) · Tailwind CSS · Node.js · Express.js · MongoDB · Mongoose · Google Gemini SDK · Groq SDK · Cloudinary

- Built a full-stack study platform that parses uploaded PDFs/notes into structured summaries, auto-generated quizzes, and a color-coded knowledge-gaps dashboard tracking topic mastery over time.
- Developed a Mock Interview Simulator that generates role- and experience-specific questions and evaluates responses in real time with scorecards, plus a Study Planner Wizard for day-by-day scheduling.
- Created an ATS Resume Reviewer performing multi-step keyword scanning and match scoring, and an Admin Console tracking API latency, token usage, and monthly LLM cost breakdowns across providers.

UniRide — Student Ride-Pooling Platform [\[Live\]](#)

Jan – Feb 2025

Node.js · Express.js · MongoDB · Socket.IO · Tailwind CSS · EJS

- Co-built a real-time ride-pooling platform helping university students find and share inter-city rides, developed during a competitive hackathon sprint.
- Implemented bidirectional WebSocket communication via Socket.IO for live ride-status updates, driver-rider matching, and instant notifications.
- Contributed across API design, EJS frontend templating, MongoDB data modelling, and deployment earning 20th Place & Public Voting Award.

EDUCATION

B.E. Software Engineering — Mehran University of Engineering & Technology, Jamshoro

Dec 2022 – Present

Relevant Coursework: Data Structures & Algorithms · Web Development · Database Systems · OOP · Software Project Management · Human-Computer Interaction · Artificial Intelligence

CERTIFICATIONS & ACHIEVEMENTS

- Fundamentals of Java Programming — Coursera (Sep 2023)
- 20th Place & Public Voting Winner — Hack For Humanity 2025, DevPost (Feb 2025)